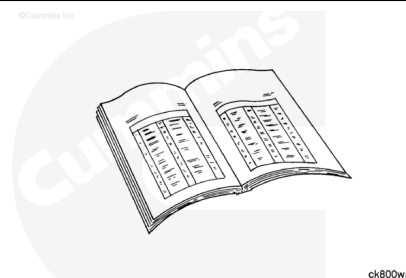


Trainings ▾

Preparatory Steps

- Remove the fan drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-002-tr.html>)
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html>)
- Remove the fan and fan hub support. Refer to Procedure 008-036 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-036-tr.html>)
- If the engine is equipped with an automatic belt tensioner, remove the belt tensioner arm from the tensioner bracket. Remove the four mounting capscrews and the tensioner bracket from the gear cover.
- Remove the accessory drive pulley. Refer to Procedure 009-004 in Section 9. (</qs3/pubsys2/xml/en/procedures/35/35-009-004-tr.html>)
- Remove the alternator drive pulley. Refer to Procedure 009-010 in Section 9. (</qs3/pubsys2/xml/en/procedures/35/35-009-010-tr.html>)
- Remove the vibration damper. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-052-tr.html>)
- Remove the front engine support. Refer to Procedure 016-002 in Section 16. (</qs3/pubsys2/xml/en/procedures/35/35-016-002-tr.html>)
- Remove the oil seals from the gear cover.
 - Refer to Procedure 001-001 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-001-tr.html>)
 - Refer to Procedure 001-003 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-003-tr.html>)



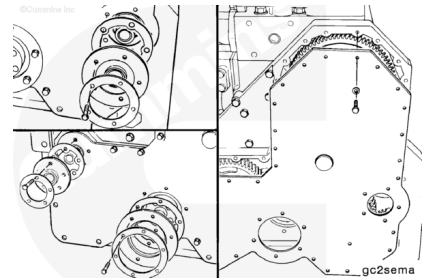
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tr.html)

- Refer to Procedure 001-023 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-023-tr.html)

Remove

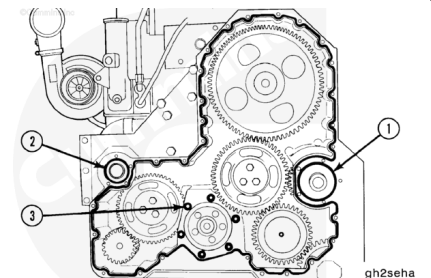
Remove the remaining capscrews and the gear cover.



Clean and Inspect for Reuse

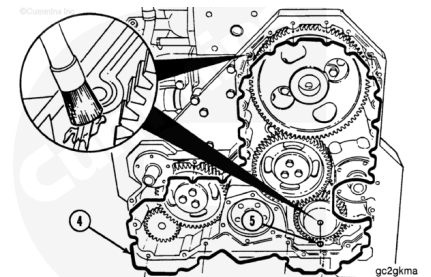
Remove the seal rings at the accessory drive mounting (1) and water pump mounting holes (2), and clean the grooves.

Remove the six o-ring seals at the crankshaft seal mounting (3), and clean the grooves.



Remove the seal (4) from the front of the gear housing, and clean the groove.

Remove the o-ring at the front of the hydraulic drive idler retainer, and clean the groove.

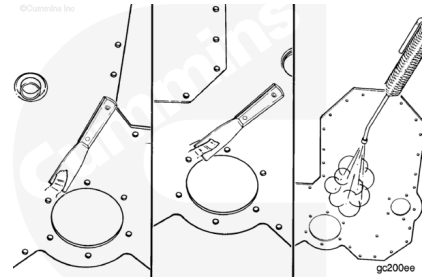


⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing, Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



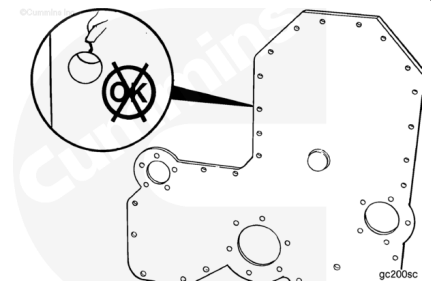
Clean the sealing surface on the rear face of the gear cover.

Clean the oil seal carrier gasket surfaces.

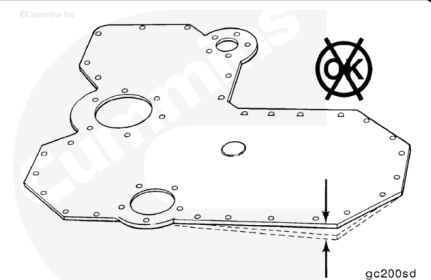
Use steam or solvent to clean the gear cover. Dry with compressed air.

Inspect the gear cover for cracks or other damage.

If the cover is cracked or otherwise damaged, it **must** be replaced.

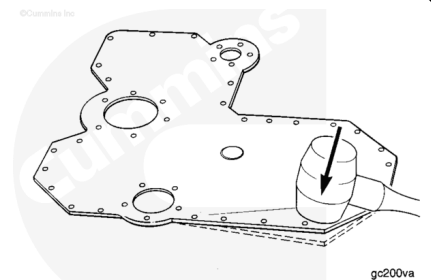


Put the gear cover on a flat surface, and inspect the gear cover for flatness.



It is an acceptable practice to use a mallet to straighten a gear cover with minor bends.

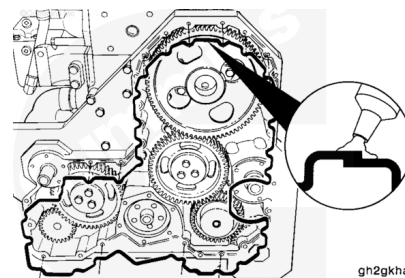
Lay the gear cover on a flat surface, and hit the cover in the bent area.



Install

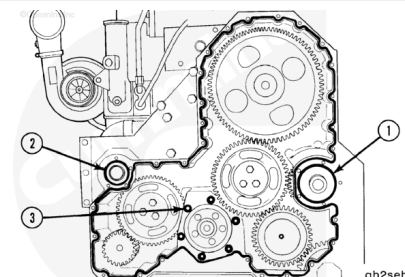
Insert a new seal into the groove on the front of the gear housing.

Apply silicone sealant, Part Number 3823494, to the seal overlap joint at the top of the gear housing.

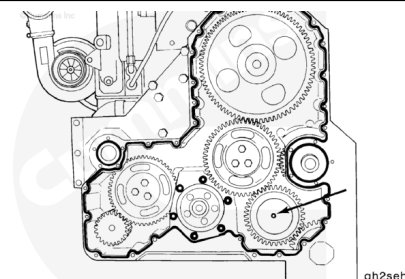


Install a new seal ring at the accessory drive mounting (1) and water pump mounting (2) locations.

Install six new capscrew o-rings at the crankshaft seal mounting capscrew locations (3).



Install a new o-ring at the front of the hydraulic drive idler retainer mounting spacer. Make sure the spacer is in place.



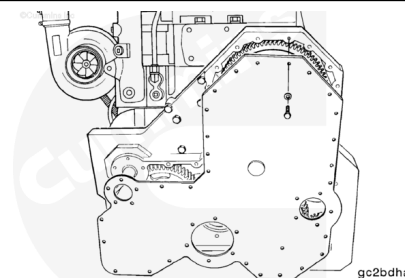
Install the gear cover and mounting capscrews.

Do **not** install the front engine support capscrews at this time.

Tighten the capscrews.

Torque Value: 6 n•m [53 in-lb]

Note : This is not the final torque value.

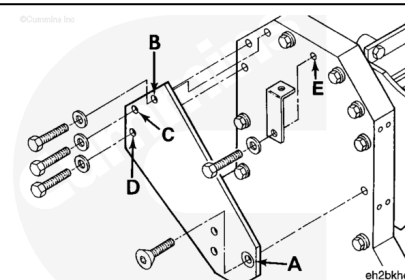


If the engine is equipped with an automatic belt tensioner, install the belt tensioner bracket with the countersunk hole (A) facing away from the gear cover. Use a flat head (M8 x 25) at location (A).

Use three capscrews (M8 x 30) and plain washers at (B, C, and D). Tighten the capscrews.

Torque Value: 6 n•m [53 in-lb]

Note : This is not the final torque value.

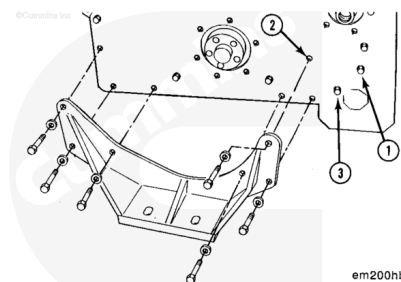


Apply a coating of thread sealant, Part Number 3823494, to the three capscrews at locations (1, 2, and 3).

Install and tighten all the front engine support and mounting capscrews to the following torque:

Torque Value: 6 n•m [53 in-lb]

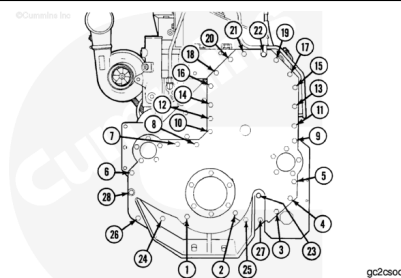
Note : This is not the final torque value.



Tighten the gear cover capscrews in the sequence shown.

Torque Value:

Capscrews 1 through 22: 20 n•m [177 in-lb]

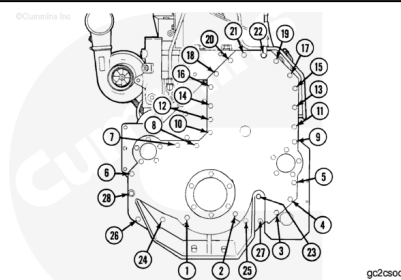


Tighten the front engine support capscrews in the sequence shown

Torque Value:

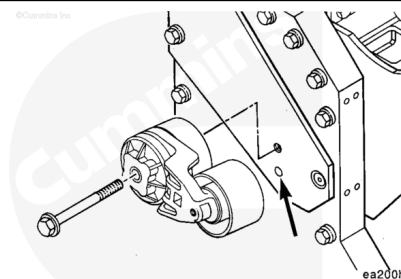
Capscrews 23 through 28: 68 n•m [50 ft-lb]

Plus 60 degrees (one flat).



Install the automatic belt tensioner with a (M10-1.50 x 70) capscrew. Align the cast locator pin on the tensioner with the hole in the bracket before tightening.

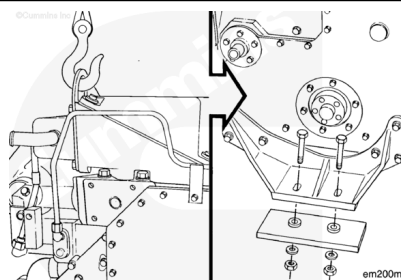
Torque Value: 43 n•m [32 ft-lb]



Lower the engine onto the crossmember.

Install the front support to crossmember.

Install the capscrews and tighten. Refer to the manufacturer's specifications for the correct torque value.



Finishing Steps

- Install the gear cover oil seals.
 - Refer to Procedure 001-001 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-001-tr.html)
 - Refer to Procedure 001-003 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-003-tr.html)
 - Refer to Procedure 001-023 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-023-tr.html)
- Install the vibration damper. Refer to Procedure 001-052 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-052-tr.html)
- Install the alternator drive pulley. Refer to Procedure 009-010 in Section 9.
(/qs3/pubsys2/xml/en/procedures/35/35-009-010-tr.html)
- Install the accessory drive pulley. Refer to Procedure 009-004 in Section 9.
(/qs3/pubsys2/xml/en/procedures/35/35-009-004-tr.html)
- Install the fan and fan hub support. Refer to Procedure 008-036 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-036-tr.html)
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13.
(/qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html)
- Install the fan drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-002-tr.html)
- Operate the engine to normal operating temperature and check for leaks.



